



IUS
INSTITUT
UNIVERSITAIRE
DES SCIENCES

INSTITUT UNIVERSITAIRE DES SCIENCES (IUS)

Faculté des sciences et des technologies (FST)

Rapport du travail de Laboratoire N°6_Systèmes

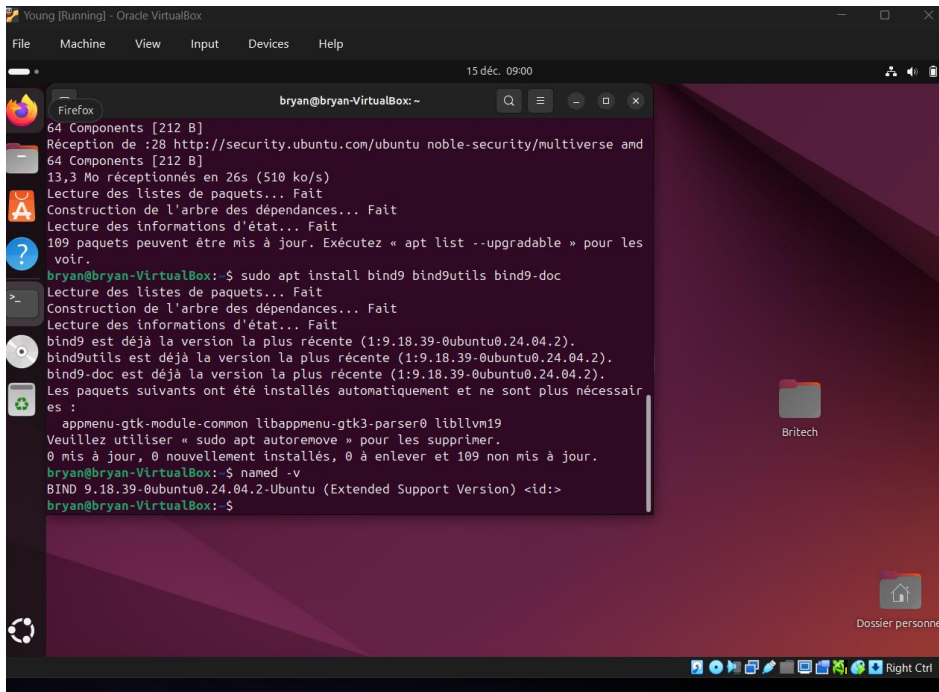
Etudiant : Brian ANTOINE

Niveau : L3

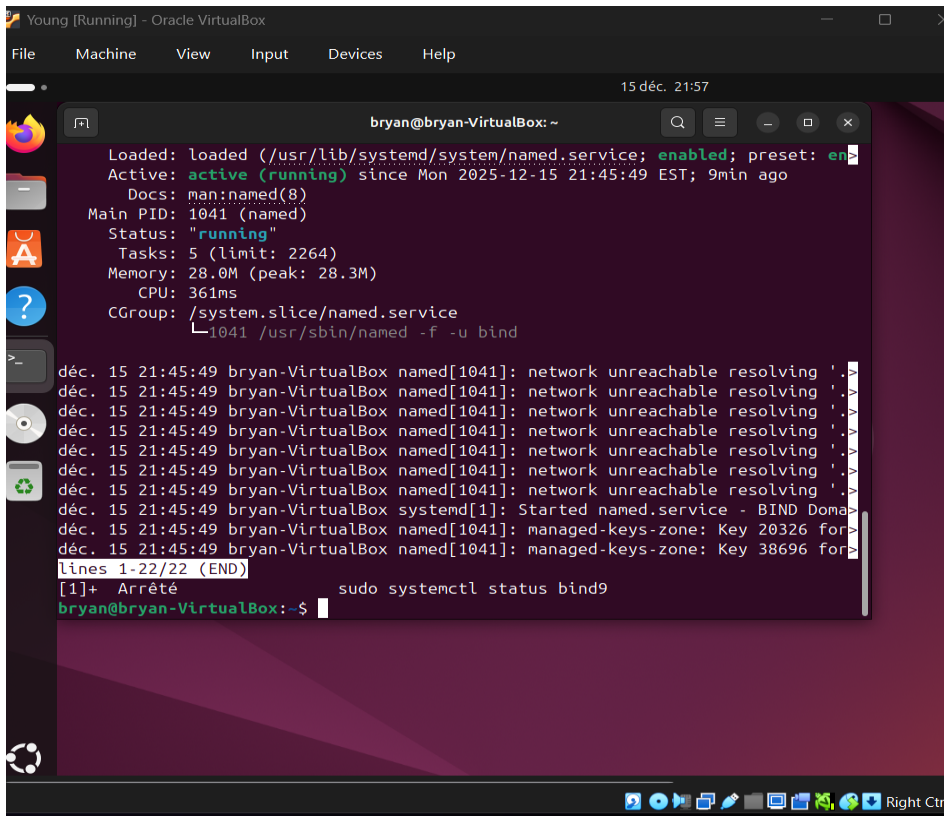
14 décembre 2025

1) Création d'un domaine interne :

➤ Installation et configuration de Bind9

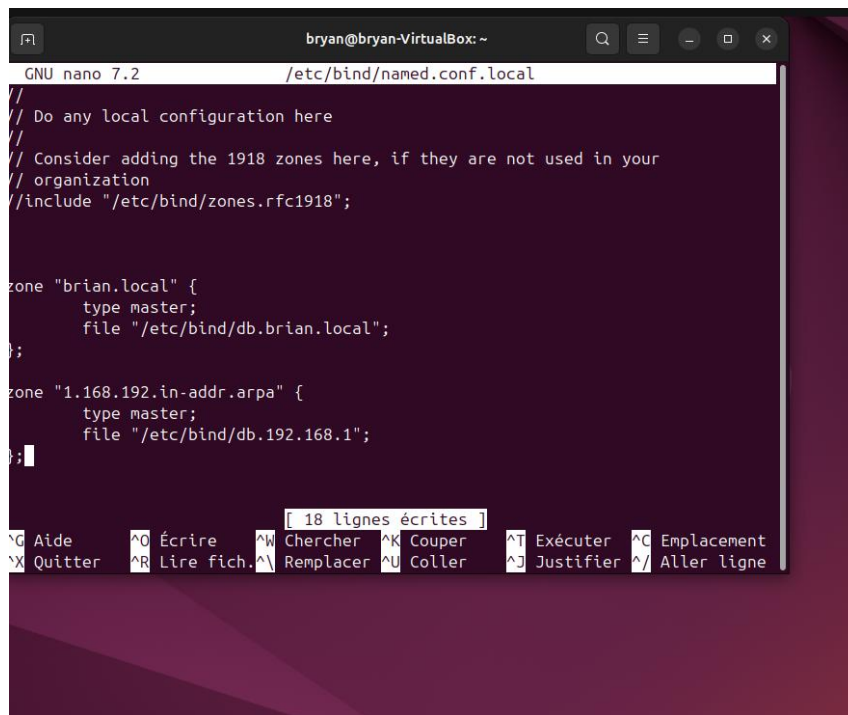


```
bryan@bryan-VirtualBox:~$ sudo apt install bind9 bind9utils bind9-doc
64 Components [212 B]
Réception de :28 http://security.ubuntu.com/ubuntu noble-security/multiverse amd
64 Components [212 B]
13,3 Mo réceptionnés en 26s (510 ko/s)
Lecture des listes de paquets... Fait
Construction de l'arbre des dépendances... Fait
Lecture des informations d'état... Fait
109 paquets peuvent être mis à jour. Exécutez « apt list --upgradable » pour les voir.
bryan@bryan-VirtualBox:~$ sudo apt install bind9 bind9utils bind9-doc
Lecture des listes de paquets... Fait
Construction de l'arbre des dépendances... Fait
Lecture des informations d'état... Fait
bind9 est déjà la version la plus récente (1:9.18.39-0ubuntu0.24.04.2).
bind9utils est déjà la version la plus récente (1:9.18.39-0ubuntu0.24.04.2).
bind9-doc est déjà la version la plus récente (1:9.18.39-0ubuntu0.24.04.2).
Les paquets suivants ont été installés automatiquement et ne sont plus nécessaires :
  appmenu-gtk-module-common libappmenu-gtk3-parser0 libllvm19
Veuillez utiliser « sudo apt autoremove » pour les supprimer.
0 mis à jour, 0 nouvellement installés, 0 à enlever et 109 non mis à jour.
bryan@bryan-VirtualBox:~$ named -v
BIND 9.18.39-0ubuntu0.24.04.2-Ubuntu (Extended Support Version) <id:>
bryan@bryan-VirtualBox:~$
```



```
bryan@bryan-VirtualBox:~$ sudo systemctl status bind9
Loaded: loaded (/usr/lib/systemd/system/named.service; enabled; preset: en
Active: active (running) since Mon 2025-12-15 21:45:49 EST; 9min ago
Docs: man:named(8)
Main PID: 1041 (named)
Status: "running"
Tasks: 5 (limit: 2264)
Memory: 28.0M (peak: 28.3M)
CPU: 361ms
CGroup: /system.slice/named.service
└─1041 /usr/sbin/named -f -u bind

déc. 15 21:45:49 bryan-VirtualBox named[1041]: network unreachable resolving '
déc. 15 21:45:49 bryan-VirtualBox named[1041]: network unreachable resolving '
déc. 15 21:45:49 bryan-VirtualBox named[1041]: network unreachable resolving '
déc. 15 21:45:49 bryan-VirtualBox named[1041]: network unreachable resolving '
déc. 15 21:45:49 bryan-VirtualBox named[1041]: network unreachable resolving '
déc. 15 21:45:49 bryan-VirtualBox named[1041]: network unreachable resolving '
déc. 15 21:45:49 bryan-VirtualBox system[1]: Started named.service - BIND Domai
déc. 15 21:45:49 bryan-VirtualBox named[1041]: managed-keys-zone: Key 20326 for
déc. 15 21:45:49 bryan-VirtualBox named[1041]: managed-keys-zone: Key 38696 for
lines 1-22/22 (END)
[1]+ Arrêté
bryan@bryan-VirtualBox:~$
```



The screenshot shows a terminal window titled "bryan@bryan-VirtualBox: ~" with the GNU nano 7.2 editor open at the file "/etc/bind/named.conf.local". The editor contains the following configuration:

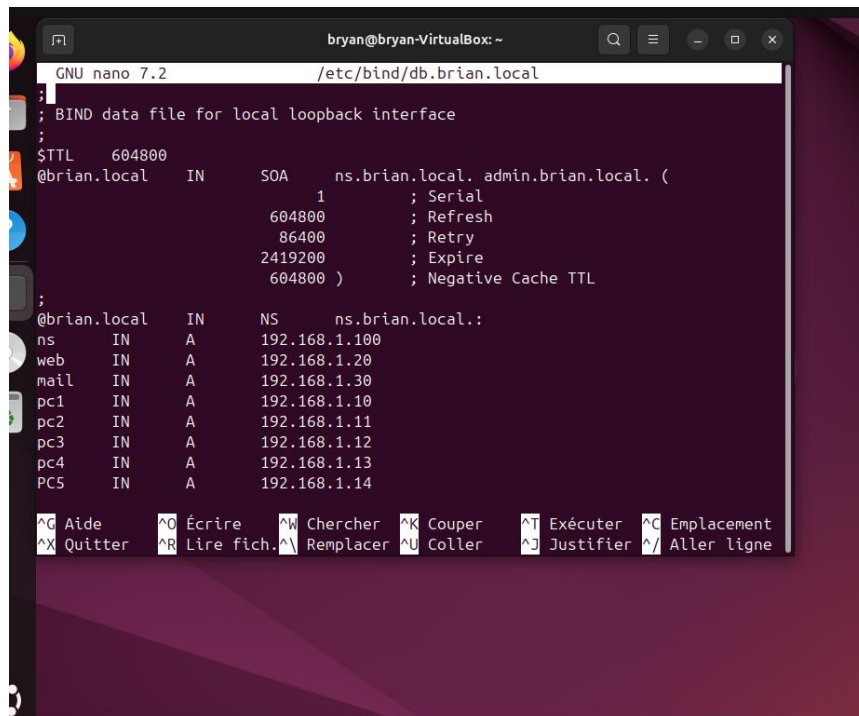
```
// Do any local configuration here
// Consider adding the 1918 zones here, if they are not used in your
// organization
//include "/etc/bind/zones.rfc1918";

zone "brian.local" {
    type master;
    file "/etc/bind/db.brian.local";
};

zone "1.168.192.in-addr.arpa" {
    type master;
    file "/etc/bind/db.192.168.1";
};
```

The bottom of the window shows a status bar with the text "[18 lignes écrites]" and a set of keyboard shortcuts: ^G Aide, ^O Écrire, ^W Chercher, ^K Couper, ^T Exécuter, ^C Emplacement, ^X Quitter, ^R Lire fich., ^_ Remplacer, ^U Coller, ^J Justifier, ^/ Aller ligne.

❖ Configuration de la Zone directe



The screenshot shows a terminal window titled "bryan@bryan-VirtualBox: ~" with the GNU nano 7.2 editor open at the file "/etc/bind/db.brian.local". The editor contains the following configuration:

```
;
; BIND data file for local loopback interface
;
$TTL 604800
@brian.local IN SOA ns.brian.local. admin.brian.local. (
    1 ; Serial
    604800 ; Refresh
    86400 ; Retry
    2419200 ; Expire
    604800 ) ; Negative Cache TTL
;
@brian.local IN NS ns.brian.local.;
ns IN A 192.168.1.100
web IN A 192.168.1.20
mail IN A 192.168.1.30
pc1 IN A 192.168.1.10
pc2 IN A 192.168.1.11
pc3 IN A 192.168.1.12
pc4 IN A 192.168.1.13
PC5 IN A 192.168.1.14
```

The bottom of the window shows a status bar with the text "[18 lignes écrites]" and a set of keyboard shortcuts: ^G Aide, ^O Écrire, ^W Chercher, ^K Couper, ^T Exécuter, ^C Emplacement, ^X Quitter, ^R Lire fich., ^_ Remplacer, ^U Coller, ^J Justifier, ^/ Aller ligne.

```
bryan@bryan-VirtualBox: ~  
Status: "running"  
Tasks: 5 (limit: 2264)  
Memory: 28.0M (peak: 28.3M)  
CPU: 361ms  
CGroup: /system.slice/named.service  
└─1041 /usr/sbin/named -f -u bind  
  
déc. 15 21:45:49 bryan-VirtualBox named[1041]: network unreachable resolving '...  
déc. 15 21:45:49 bryan-VirtualBox named[1041]: network unreachable resolving '...  
déc. 15 21:45:49 bryan-VirtualBox named[1041]: network unreachable resolving '...  
déc. 15 21:45:49 bryan-VirtualBox named[1041]: network unreachable resolving '...  
déc. 15 21:45:49 bryan-VirtualBox named[1041]: network unreachable resolving '...  
déc. 15 21:45:49 bryan-VirtualBox named[1041]: network unreachable resolving '...  
déc. 15 21:45:49 bryan-VirtualBox systemd[1]: Started named.service - BIND Domain  
déc. 15 21:45:49 bryan-VirtualBox named[1041]: managed-keys-zone: Key 20326 for  
déc. 15 21:45:49 bryan-VirtualBox named[1041]: managed-keys-zone: Key 38696 for  
lines 1-22/22 (END)  
[1]+ Arrêté  
bryan@bryan-VirtualBox:~$ sudo systemctl status bind9  
bryan@bryan-VirtualBox:~$ sudo nano /etc/bind/named.conf.local  
bryan@bryan-VirtualBox:~$ sudo cp /etc/bind/db.local /etc/bind/db.brian.local  
bryan@bryan-VirtualBox:~$ sudo nano /etc/bind/db.brian.local  
bryan@bryan-VirtualBox:~$ sudo cp /etc/bind/db.127 /etc/bind/db.192.168.1  
bryan@bryan-VirtualBox:~$ sudo nano /etc/bind/db.192.168.1
```

❖ Configuration de la Zone inverse

```
GNU nano 7.2 /etc/bind/db.192.168.1  
; BIND reverse data file for local loopback interface  
;  
$TTL 604800  
@ IN SOA ns.brian.local. admin.brian.local. (  
1 ; Serial  
604800 ; Refresh  
86400 ; Retry  
2419200 ; Expire  
604800 ) ; Negative Cache TTL  
;  
@ IN NS ns.brian.local.  
100 IN PTR ns.brian.local.  
20 IN PTR web.brian.local.  
30 IN PTR mail.brian.local.  
10 IN PTR pc1.brian.local.  
11 IN PTR pc2.brian.local.  
12 IN PTR pc3.brian.local.  
13 IN PTR pc4.brian.local.  
14 IN PTR pc5.brian.local.  
[ 20 lignes écrites ]  
^G Aide ^O Écrire ^W Chercher ^K Couper ^T Exécuter ^C Emplacement  
^X Quitter ^R Lire fich. ^_ Remplacer ^U Coller ^J Justifier ^_ Aller ligne
```

```
15 déc. 22:29
bryan@bryan-VirtualBox: ~
déc. 15 21:45:49 bryan-VirtualBox named[1041]: managed-keys-zone: Key 38696 for>
lines 1-22/22 (END)
Fichiers  rêté      sudo systemctl status bind9
bryan@bryan-VirtualBox:~$ sudo nano /etc/bind/named.conf.local
bryan@bryan-VirtualBox:~$ sudo cp /etc/bind/db.local /etc/bind/db.brian.local
bryan@bryan-VirtualBox:~$ sudo nano /etc/bind/db.brian.local
bryan@bryan-VirtualBox:~$ sudo cp /etc/bind/db.127 /etc/bind/db.192.168.1
bryan@bryan-VirtualBox:~$ sudo nano /etc/bind/db.192.168.1
bryan@bryan-VirtualBox:~$ sudo named-checkconf
bryan@bryan-VirtualBox:~$ sudo named-checkzone brian.local /etc/bind/db.brian.lo
cal
/etc/bind/db.brian.local:5: SOA record not at top of zone (@brian.local.brian.l
ocal)
/etc/bind/db.brian.local:12: warning: ns.brian.local.:brian.local: bad name (ch
eck-names)
zone brian.local/IN: loading from master file /etc/bind/db.brian.local failed: n
ot at top of zone
zone brian.local/IN: not loaded due to errors.
bryan@bryan-VirtualBox:~$ sudo named-checkzone 1.168.192.in-addr.arpa /etc/bind/
db.192.168.1
zone 1.168.192.in-addr.arpa/IN: loaded serial 1
OK
bryan@bryan-VirtualBox:~$ sudo systemctl restart bind9
bryan@bryan-VirtualBox:~$ sudo systemctl status bind9
```

❖ Tests de Validation

```
bryan@bryan-VirtualBox: ~
déc. 15 21:45:49 bryan-VirtualBox named[1041]: managed-keys-zone: Key 20326 for>
déc. 15 21:45:49 bryan-VirtualBox named[1041]: managed-keys-zone: Key 38696 for>
lines 1-22/22 (END)
Fichiers  Arrêté      sudo systemctl status bind9
bryan@bryan-VirtualBox:~$ sudo nano /etc/bind/named.conf.local
bryan@bryan-VirtualBox:~$ sudo cp /etc/bind/db.local /etc/bind/db.brian.local
bryan@bryan-VirtualBox:~$ sudo nano /etc/bind/db.brian.local
bryan@bryan-VirtualBox:~$ sudo cp /etc/bind/db.127 /etc/bind/db.192.168.1
bryan@bryan-VirtualBox:~$ sudo nano /etc/bind/db.192.168.1
bryan@bryan-VirtualBox:~$ sudo named-checkconf
bryan@bryan-VirtualBox:~$ sudo named-checkzone brian.local /etc/bind/db.brian.lo
cal
/etc/bind/db.brian.local:5: SOA record not at top of zone (@brian.local.brian.l
ocal)
/etc/bind/db.brian.local:12: warning: ns.brian.local.:brian.local: bad name (ch
eck-names)
zone brian.local/IN: loading from master file /etc/bind/db.brian.local failed: n
ot at top of zone
zone brian.local/IN: not loaded due to errors.
bryan@bryan-VirtualBox:~$ sudo named-checkzone 1.168.192.in-addr.arpa /etc/bind/
db.192.168.1
zone 1.168.192.in-addr.arpa/IN: loaded serial 1
OK
```

```
Young [running] - Oracle VM VirtualBox
File Machine View Input Devices Help
15 déc. 22:34
bryan@bryan-VirtualBox: ~
bryan@bryan-VirtualBox:~$ dig brian.local 192.168.1.100

; <<>> DiG 9.18.39-0ubuntu0.24.04.2-Ubuntu <<>> brian.local 192.168.1.100
;; global options: +cmd
;; Got answer:
;; WARNING: .local is reserved for Multicast DNS
;; You are currently testing what happens when an mDNS query is leaked to DNS
;; ->>HEADER<<- opcode: QUERY, status: SERVFAIL, id: 52728
;; flags: qr aa rd ra; QUERY: 1, ANSWER: 0, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
;; EDNS: version: 0, flags:; udp: 65494
;; QUESTION SECTION:
;brian.local.                IN      A

;; Query time: 0 msec
;; SERVER: 127.0.0.53#53(127.0.0.53) (UDP)
;; WHEN: Mon Dec 15 22:34:05 EST 2025
;; MSG SIZE rcvd: 40

;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 52830
;; flags: qr rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 1
```

```
bryan@bryan-VirtualBox: ~
bryan@bryan-VirtualBox:~$ dig -x 192.168.1.20

; <<>> DiG 9.18.39-0ubuntu0.24.04.2-Ubuntu <<>> -x 192.168.1.20
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NXDOMAIN, id: 5593
;; flags: qr rd ra; QUERY: 1, ANSWER: 0, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
;; EDNS: version: 0, flags:; udp: 65494
;; QUESTION SECTION:
;20.1.168.192.in-addr.arpa.  IN      PTR

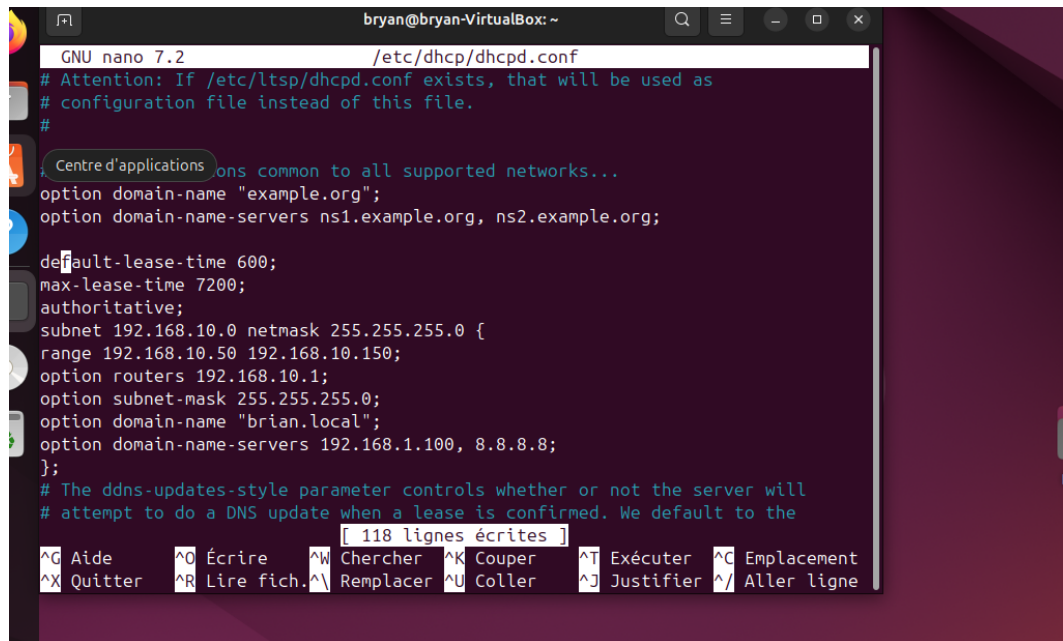
;; Query time: 141 msec
;; SERVER: 127.0.0.53#53(127.0.0.53) (UDP)
;; WHEN: Mon Dec 15 22:37:04 EST 2025
;; MSG SIZE rcvd: 54
```


➤ Configuration d'un serveur DHCP et analyse des logs :

```
bryan@bryan-VirtualBox: ~  
bryan@bryan-VirtualBox:~$ sudo apt install isc-dhcp-server  
[sudo] Mot de passe de bryan :  
( Fichiers des listes de paquets... Fait  
Construction de l'arbre des dépendances... Fait  
Lecture des informations d'état... Fait  
Les paquets suivants ont été installés automatiquement et ne sont plus nécessair  
es :  
  appmenu-gtk-module-common libappmenu-gtk3-parser0 libllvm19  
Veuillez utiliser « sudo apt autoremove » pour les supprimer.  
Les paquets supplémentaires suivants seront installés :  
  isc-dhcp-common  
Paquets suggérés :  
  isc-dhcp-server-ldap policycoreutils  
Les NOUVEAUX paquets suivants seront installés :  
  isc-dhcp-common isc-dhcp-server  
0 mis à jour, 2 nouvellement installés, 0 à enlever et 113 non mis à jour.  
Il est nécessaire de prendre 1 281 ko dans les archives.  
Après cette opération, 4 281 ko d'espace disque supplémentaires seront utilisés.  
Souhaitez-vous continuer ? [0/n] o  
Réception de :1 http://archive.ubuntu.com/ubuntu noble/universe amd64 isc-dhcp-s  
erver amd64 4.4.3-P1-4ubuntu2 [1 236 kB]  
Réception de :2 http://archive.ubuntu.com/ubuntu noble/universe amd64 isc-dhcp-c  
ommon amd64 4.4.3-P1-4ubuntu2 [45,8 kB]  
1 281 ko réceptionnés en 6s (217 ko/s)
```

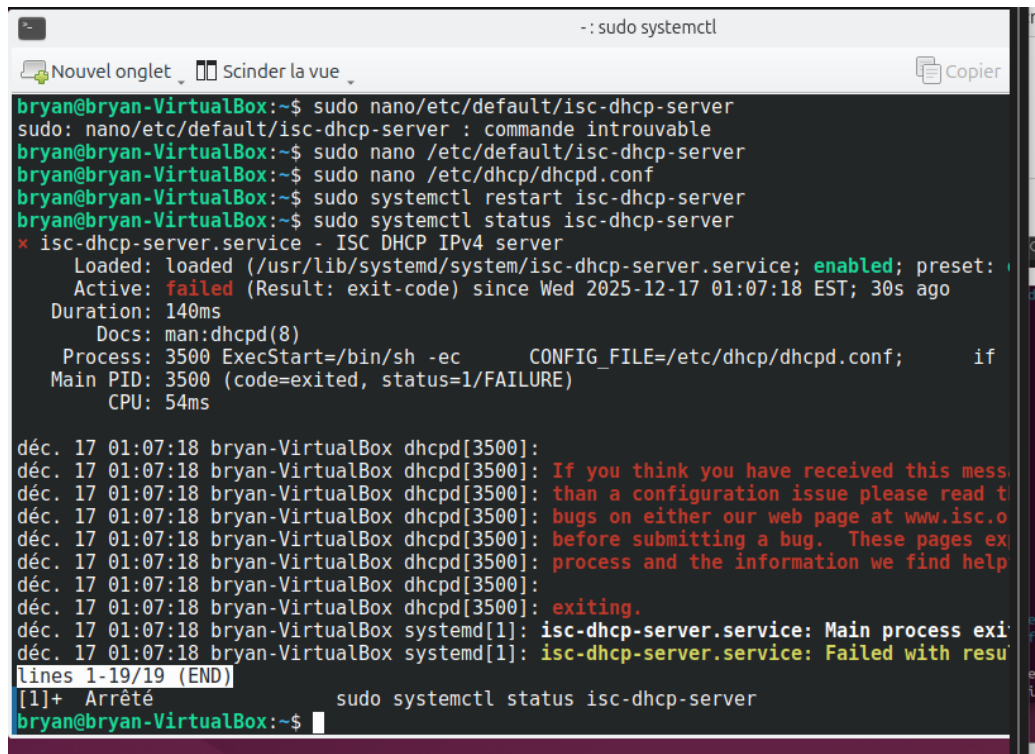
```
bryan@bryan-VirtualBox: ~  
ommon amd64 4.4.3-P1-4ubuntu2 [45,8 kB]  
1 281 ko réceptionnés en 6s (217 ko/s)  
Préconfiguration des paquets...  
Sélection du paquet isc-dhcp-server précédemment désélectionné.  
(Lecture de la base de données... 273638 fichiers et répertoires déjà installés.  
)  
Préparation du dépaquetage de .../isc-dhcp-server_4.4.3-P1-4ubuntu2_amd64.deb ..  
Dépaquetage de isc-dhcp-server (4.4.3-P1-4ubuntu2) ...  
Sélection du paquet isc-dhcp-common précédemment désélectionné.  
Préparation du dépaquetage de .../isc-dhcp-common_4.4.3-P1-4ubuntu2_amd64.deb ..  
Dépaquetage de isc-dhcp-common (4.4.3-P1-4ubuntu2) ...  
Paramétrage de isc-dhcp-server (4.4.3-P1-4ubuntu2) ...  
Generating /etc/default/isc-dhcp-server...  
Created symlink /etc/systemd/system/multi-user.target.wants/isc-dhcp-server.serv  
ice → /usr/lib/systemd/system/isc-dhcp-server.service.  
Created symlink /etc/systemd/system/multi-user.target.wants/isc-dhcp-server6.ser  
vice → /usr/lib/systemd/system/isc-dhcp-server6.service.  
Paramétrage de isc-dhcp-common (4.4.3-P1-4ubuntu2) ...  
Traitement des actions différées (« triggers ») pour man-db (2.12.0-4build2) ...  
bryan@bryan-VirtualBox:~$
```

- Configuration



```
bryan@bryan-VirtualBox: ~  
GNU nano 7.2 /etc/dhcp/dhcpd.conf  
# Attention: If /etc/ltsp/dhcpd.conf exists, that will be used as  
# configuration file instead of this file.  
#  
# Centre d'applications  
# Options common to all supported networks...  
option domain-name "example.org";  
option domain-name-servers ns1.example.org, ns2.example.org;  
  
default-lease-time 600;  
max-lease-time 7200;  
authoritative;  
subnet 192.168.10.0 netmask 255.255.255.0 {  
  range 192.168.10.50 192.168.10.150;  
  option routers 192.168.10.1;  
  option subnet-mask 255.255.255.0;  
  option domain-name "brian.local";  
  option domain-name-servers 192.168.1.100, 8.8.8.8;  
};  
# The ddns-updates-style parameter controls whether or not the server will  
# attempt to do a DNS update when a lease is confirmed. We default to the  
# none mode for now.  
#ddns-updates-style none;  
[ 118 lignes écrites ]  
^G Aide      ^O Écrire    ^W Chercher  ^K Couper    ^T Exécuter  ^C Emplacement  
^X Quitter   ^R Lire fich.^N Remplacer  ^U Coller    ^J Justifier ^_ Aller ligne
```

- Problème rencontré avec le serveur DHCP



```
- : sudo systemctl  
Nouvel onglet Scinder la vue Copier  
bryan@bryan-VirtualBox:~$ sudo nano/etc/default/isc-dhcp-server  
sudo: nano/etc/default/isc-dhcp-server : commande introuvable  
bryan@bryan-VirtualBox:~$ sudo nano /etc/default/isc-dhcp-server  
bryan@bryan-VirtualBox:~$ sudo nano /etc/dhcp/dhcpd.conf  
bryan@bryan-VirtualBox:~$ sudo systemctl restart isc-dhcp-server  
bryan@bryan-VirtualBox:~$ sudo systemctl status isc-dhcp-server  
* isc-dhcp-server.service - ISC DHCP IPv4 server  
   Loaded: loaded (/usr/lib/systemd/system/isc-dhcp-server.service; enabled; preset: enabled)  
   Active: failed (Result: exit-code) since Wed 2025-12-17 01:07:18 EST; 30s ago  
     Duration: 140ms  
       Docs: man:dhcpd(8)  
    Process: 3500 ExecStart=/bin/sh -ec CONFIG_FILE=/etc/dhcp/dhcpd.conf; if [ -f $CONFIG_FILE ]; then dhcpd -cf $CONFIG_FILE; else dhcpd; fi; && [ $? = 0 ] && exit 0;  
   Main PID: 3500 (code=exited, status=1/FAILURE)  
      CPU: 54ms  
  
déc. 17 01:07:18 bryan-VirtualBox dhcpd[3500]:  
déc. 17 01:07:18 bryan-VirtualBox dhcpd[3500]: If you think you have received this message,  
déc. 17 01:07:18 bryan-VirtualBox dhcpd[3500]: then a configuration issue please read the  
déc. 17 01:07:18 bryan-VirtualBox dhcpd[3500]: bugs on either our web page at www.isc.org  
déc. 17 01:07:18 bryan-VirtualBox dhcpd[3500]: before submitting a bug. These pages explain the  
déc. 17 01:07:18 bryan-VirtualBox dhcpd[3500]: process and the information we find helpful.  
déc. 17 01:07:18 bryan-VirtualBox dhcpd[3500]: exiting.  
déc. 17 01:07:18 bryan-VirtualBox systemd[1]: isc-dhcp-server.service: Main process exited, code=exited, status=1/FAILURE  
déc. 17 01:07:18 bryan-VirtualBox systemd[1]: isc-dhcp-server.service: Failed with result 'exit-code'.  
lines 1-19/19 (END)  
[1]+ Arrêté  
bryan@bryan-VirtualBox:~$ sudo systemctl status isc-dhcp-server
```



```
~:journalctl
Nouvel onglet Scinder la vue Copier
For info, please visit https://www.isc.org/software/dhcp/
/etc/dhcp/dhcpd.conf line 22: expecting a declaration
};
^
Configuration file errors encountered -- exiting

If you think you have received this message due to a bug rather
than a configuration issue please read the section on submitting
bugs on either our web page at www.isc.org or in the README file
before submitting a bug. These pages explain the proper
process and the information we find helpful for debugging.

exiting.
bryan@bryan-VirtualBox:~$ journalctl -u isc-dhcp-server -f
déc. 17 01:07:18 bryan-VirtualBox dhcpd[3500]:
déc. 17 01:07:18 bryan-VirtualBox dhcpd[3500]: If you think you have received this mess
g
déc. 17 01:07:18 bryan-VirtualBox dhcpd[3500]: than a configuration issue please read t
e
déc. 17 01:07:18 bryan-VirtualBox dhcpd[3500]: bugs on either our web page at www.isc.o
e
déc. 17 01:07:18 bryan-VirtualBox dhcpd[3500]: before submitting a bug. These pages ex
déc. 17 01:07:18 bryan-VirtualBox dhcpd[3500]: process and the information we find help
déc. 17 01:07:18 bryan-VirtualBox dhcpd[3500]:
déc. 17 01:07:18 bryan-VirtualBox dhcpd[3500]: exiting.
déc. 17 01:07:18 bryan-VirtualBox systemd[1]: isc-dhcp-server.service: Main process exi
s=1/FAILURE
déc. 17 01:07:18 bryan-VirtualBox systemd[1]: isc-dhcp-server.service: Failed with resu
```